HAXLR8 3.0

NATIONAL LEVEL HACKATHON • MIT MYSORE

AGRICULTURE

PROBLEM STATEMENTS

15 Engineering & ML Challenges | Pick one. Build it. Grow the future.

Agri-tech beautifully intertwines engineering and farming, using robotics, sensing, and machine learning to turn everyday field challenges in crops, soil, water, livestock, and waste into smarter, more sustainable harvests.

1

Autonomous Robotic Fruit Harvesting System

Develop a robotic system that identifies ripe fruits, calculates their position and orientation, and harvests them without damaging the fruit or plant. It should use computer vision, robotic movement, and precision gripping.

2

Underground Soil Intelligence and Root Monitoring System

Build a sensor-based system that monitors underground root-zone conditions, including moisture distribution, soil compaction, temperature, and root development, to identify problems that cannot be seen from the surface.

3

Smart Livestock Health Monitoring and Early Warning System

Develop a wearable device for cattle that monitors body temperature, movement, feeding behaviour, and activity patterns to identify abnormal behaviour and provide early health alerts.

4

Autonomous Agricultural Weed Removal Robot

Design a small field robot that travels between crop rows, identifies unwanted plants, and physically removes them using a precision mechanical mechanism without damaging nearby crops.

5

Intelligent Grain Storage and Self-Protection System

Develop a smart grain-storage unit that detects internal temperature variations, moisture accumulation, insect activity, and possible fungal growth, then automatically activates ventilation or other protective mechanisms.

6

Portable AI-Based Fruit Quality Scanner

Build a portable scanning device that uses optical or near-infrared sensing to estimate internal fruit quality, such as ripeness, sweetness, and hidden defects, without cutting the fruit.

7

Autonomous Agricultural Pipeline Leak Detection Robot

Develop a robotic inspection system that detects underground or hidden irrigation pipeline leaks using acoustic, pressure, or vibration sensors and identifies the approximate leak location.

8

Solar-Powered Atmospheric Water Harvesting System

Design a prototype that extracts water from atmospheric humidity using a solar-powered condensation or desiccant-based mechanism, with automatic water collection and efficiency monitoring.

9

Smart Agricultural Waste-to-Biochar Conversion System

Develop a controlled prototype that converts agricultural residue into biochar while monitoring temperature, oxygen, and processing conditions to improve output quality and reduce harmful emissions.

10

AI-Based Seed Genetic Purity and Variety Identification

Develop and train a machine-learning model to identify crop seed varieties and detect possible genetic or physical admixture using seed images, spectral signatures, and laboratory reference data.

11

Real-Time Crop Pollination Efficiency Prediction

Build an AI model that analyses bee movement, flower density, environmental conditions, and pollinator activity from video streams to estimate pollination efficiency and predict potential fruit-set problems.

12

AI-Based Agricultural Chemical Residue Estimation

Train an ML model using spectral measurements and laboratory-verified pesticide residue datasets to estimate residue levels in agricultural produce and identify samples requiring laboratory testing.

13

Deep Learning-Based Plant Growth Stage and Development Prediction

Develop a model that learns from continuous plant images and environmental data to identify growth stages and predict future developmental milestones such as flowering, fruit formation, and maturity.

14

AI-Based Microplastic Detection in Agricultural Soil

Develop a computer-vision and ML model that identifies and classifies microplastic particles in soil microscopy images, distinguishing them from organic matter and other soil particles.

15

Machine Learning-Based Agricultural Biogas Production Optimization

Train a predictive model using organic waste composition, moisture, temperature, pH, and bioreactor operating data to estimate biogas production and identify optimal digestion conditions.

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Select your problem statement and register for HAXLR8 3.0

